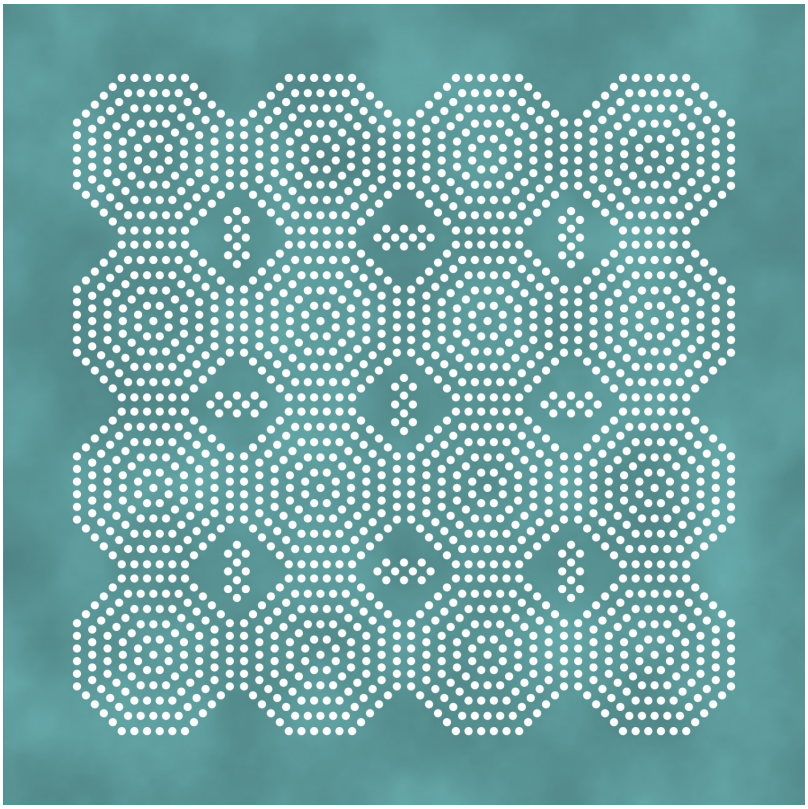
Picture for 2026



Centered Octogonal Numbers

$$
\begin{aligned}
O_1 &= 1 \\
O_2 &= 1 + 8 = 9 \\
O_3 &= 1 + 8 + 16 = 25 \\
&\vdots \\
O_n &= 1 + 8 + 16 + \cdots + 8\,(n-1) \\
&= 1 + 8\,(1 + 2 + \cdots + (n-1)) \\
&= 1 + 8\left(\frac{(n-1)\,(n)}{2}\right) \\
&= 1 + 4\,(n-1)\,(n) \\
&= 1 + 4n^2 - 4n
\end{aligned}
$$

Now, $O_6 = 121$, and $16\,(121) + 9\,(10) = 2026$.

Also, $2026 = 2\,(1013)$, so 2026 is the product of two primes.

Cantor Set

Geometrically, start with the unit interval.

$$C_0 = [0, 1] = \{x \mid 0 \leq x \leq 1\}$$

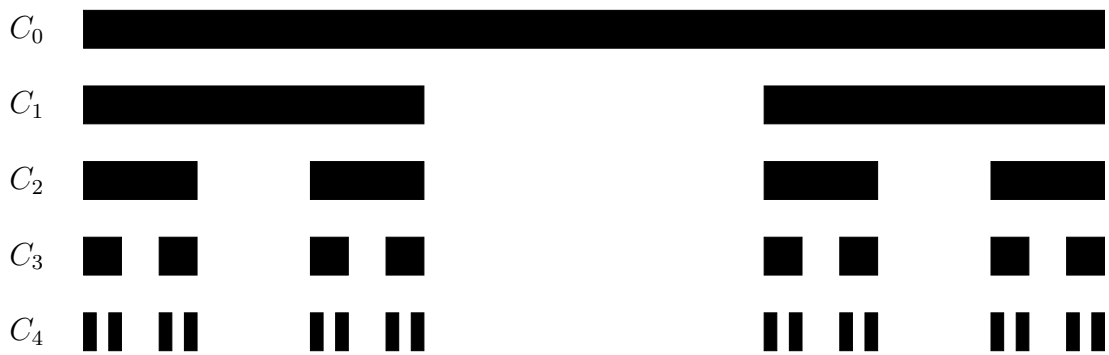
Rule for iteration: Remove the middle third

$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] = \left\{x \mid 0 \leq x \leq \frac{1}{3} \text{ or } \frac{2}{3} \leq x \leq 1\right\}$$

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]$$

$$C_3 = \left[0, \frac{1}{27}\right] \cup \left[\frac{2}{27}, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{7}{27}\right] \cup \left[\frac{8}{27}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{19}{27}\right] \cup \left[\frac{20}{27}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, \frac{25}{27}\right] \cup \left[\frac{26}{27}, 1\right]$$

Graphically, this looks like:



Repeat this process ad infinitum.

C_n consists of 2^{n-1} closed intervals of length $\left(\frac{1}{3}\right)^{n-1}$.

Cantor Set: $C = \bigcap_{n=1}^{\infty} C_n$, and has total length 0.

Base 3 (Ternary Numbers)

- The possible digits are 0, 1, 2

Decimal	Ternary
0	0
1	1
2	2
3	10
4	11
5	12
6	20
7	21
8	22
9	100
10	101
1/3	0.1
2/3	0.2
1/9	0.01
2/9	0.02
7/9	0.21
8/9	0.22

The Cantor Set consists of all points of the form:

$$0.c_1c_2c_3\ldots \qquad \text{where } c_i \in \{0,2\}$$

What about $\left(\frac{1}{3}\right)_{10} = 0.1_3$?

Consider $0.0\overline{222}_3$, and convert it to decimal

$$0\left(\frac{1}{3}\right) + 2\left(\frac{1}{9}\right) + 2\left(\frac{1}{27}\right) + \cdots = \frac{2}{9}\left(1 + \frac{1}{3} + \frac{1}{3^2} + \ldots\right) = \frac{2}{9}\left(\frac{1}{1-\frac{1}{3}}\right) = \frac{1}{3}$$

The idea is removing middle thirds if equivalent to saying that we are keeping only the numbers that have digits that are only 0's or 2's.

It can be helpful to have different ways to represent a set depending on what we are trying to do with it.

Claim: The Cantor Set C is uncountable.

Proof. Suppose the Cantor Set is countable. Now, list all members of the set:

$$a_1 = 0.a_{11}a_{12}\ldots$$

$$a_2 = 0.a_{21}a_{22}\ldots$$

Where each $a_{ij} \in \{0,2\}$.

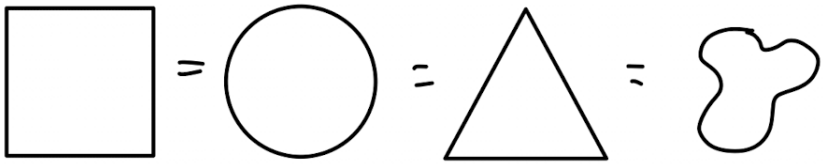
Now, let $b = 0.b_1b_2\ldots$ where $b_i = \begin{cases} 0 & \text{if } a_{ii} = 2 \\ 1 & \text{if } a_{ii} = 0 \end{cases}$.

This is an element of C that is not on the list, thus C is not countable. $\square$

Introduction to Topology

The rough idea of topology is rubber sheet geometry. We aren't concerned with rigid notions such as distance. Instead we are concerned with properties like the number of holes, and how connected an object is.

Intuitive Idea about Topology : 2 objects are topologically equivalent if you can transform one to the other without breaking, making new holes, etc.



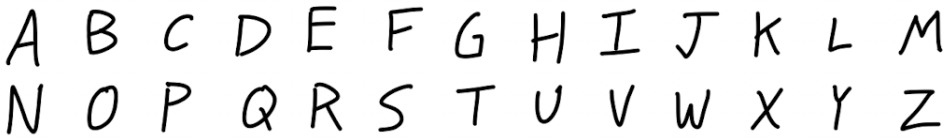
These are all examples of simple closed curves

Theorem : Jordan Curve Theorem

A simple closed curve divides the plane into an inside and an outside

Topological Classes

Consider the following alphabet (Note : the way each letter is drawn in important)



The topological classes are:

1. Simple closed curves: D, O
2. Simple open curves: C, G, L, M, N, S, U, V, W, Z
3. Simple closed curve with 2 simple open curves coming out of 2 different points: A, R
4. Q
5. B
6. E, F, J, T, Y
7. H, I, K
8. P,
9. X

- One way to check is to see what happens when you remove a point

Topology in $\mathbb{R}$

(We will be using the usual Euclidean metric on $\mathbb{R}$ which says the distance between two numbers x and y is $|x-y|$)

For a general metric space (a space X with some distance on it) we can generalize $|x-y|$ to $d(P,Q)$, the distance between points P and Q in X.

Definition : Delta Neighbourhood

Given $\delta > 0$, the δ-neighbourhood of a point $x \in \mathbb{R}$ is denoted by $V_\delta(x) = (x-\delta, x+\delta) = \{y \in \mathbb{R} : |x-y| < \delta\}$

Example

Let $X = \mathbb{R}^2$, and use the Euclidean metric: $d(P,Q) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.

$V_\delta(P) = \{Q \in \mathbb{R}^2 : d(P,Q) < \delta\}$

What is $d(P,Q) = \delta$?

$$\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2} = \delta \Rightarrow (x_2-x_1)^2 + (y_2-y_1)^2 - \delta^2$$

Let $P = (1,2)$, $\delta = 3 \Rightarrow (x_2-1)^2 + (y_2-2)^2 = 3^2$

So, $V_\delta(P)$ is the interior of the circle.

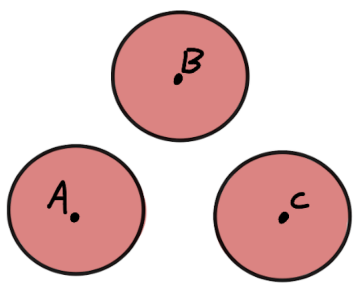
Consider an equilateral triangle (where each side has length 1), and let $S = \{A, B, C\}$, where A, B and C are the vertices of the triangle.

$$V_\delta(S) = \bigcup_{x \in S} V_\delta(x)$$

(The δ-neighbourhood of a set is the union of the δ-neighbourhoods of the points in the set

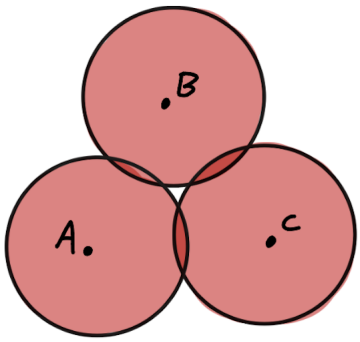
What do these δ-neighbourhoods look like? (It depends on the value of δ)

- $\delta < \frac{1}{2}$



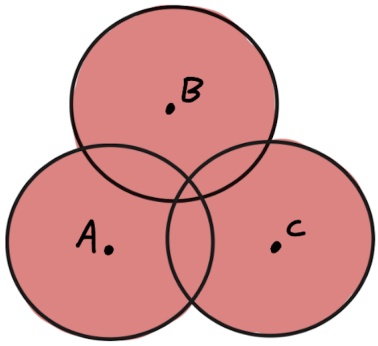
  This gives us 3 disjoint open sets

- $\delta > \frac{1}{2}$ (but not too big)



Here, the center point will not be covered

- $\delta > \frac{\frac{1}{2}}{\cos(30°)}$



Here, there will be no holes

Open Sets

Definition : Open Set

A set $U \subseteq \mathbb{R}$ is open if for every $x \in U$, there exists a $\delta > 0$ such that $V_\delta(x) = (x - \delta, x + \delta) \subseteq U$

Examples of open sets:

1. $\mathbb{R}$ itself, for any $x \in \mathbb{R}$, you can take any $\delta > 0$ and $V_\delta(x) \subseteq \mathbb{R}$
2. The empty set $\emptyset$, there are no elements in the set to verify the condition
3. $\mathbb{Q}$ is not open, given any $x \in \mathbb{Q}$ and any $\delta > 0$, the set $(x - \delta, x + \delta)$ contains irrationals
4. Any open interval (a, b), let $x \in (a, b)$, let $\delta = \min\{|x - a|, |x - b|\}$, then $V_\delta(x) \subseteq (a, b)$
5. (a, ∞) and $(-\infty, b)$ are both open for any $a, b \in \mathbb{R}$, $(a, \infty) \to \delta = |x - a|$, $(-\infty, b) \to \delta = |x - b|$
6. $[a, b]$ is not open, consider $x = a$, let $\delta > 0$, then $(a - \delta, a + \delta)$ contains points of $\mathbb{R}$ not in $[a, b]$ for example, $a - \frac{\delta}{2} \in (a - \delta, a + \delta)$, but $a - \frac{\delta}{2} < a$ so $1 - \frac{\delta}{2} \notin [a, b]$

Collections of Sets

- Finite : $\{U_1, U_2, \dots, U_n\}$ $(I = \{1, 2, \dots n\})$
- Countably Infinite : $\{U_1, U_2, \dots\} = \{U_n : n \in \mathbb{N}\}$ $(I = \mathbb{N})$
- The index set for a collection of sets is a way to describe the collection (we can use any set as an index set, for example $I = \mathbb{R}$, $I = \mathcal{P}(\mathbb{R})$, etc.)

Proposition

1. If $\{U_\alpha\}$ is any collection of open sets, then $\bigcup\limits_\alpha U_\alpha$ is open
2. If $\{U_1, U_2, \dots, U_n\}$ is a finite collection of open sets, then $\bigcap\limits_{k=1}^{n} U_k$ is open

We can do arbitrary unions and finite intersections of open sets, and end up with an open set

Proof.

1. Let I be an index set for an arbitrary collection of open sets $\{U_\alpha : \alpha \in I\}$. Let $x \in \bigcup_{\alpha \in I} U_\alpha$, there exists $\alpha_0 \in I$ such that $x \in U_{\alpha_0}$, U_{α_0} is open so there exists $\delta > 0$ such that $x \in V_\delta(x) \subseteq U_{\alpha_0}$. We know from set theory that $U_{\alpha_0} \subseteq \bigcup_{\alpha \in I} U_\alpha$, thus $V_\delta(x) \subseteq U_{\alpha_0} \subseteq \bigcup_{\alpha \in I} U_\alpha$, which proves that $\bigcup_{\alpha \in I} U_\alpha$ is open
2. Let $\{U_1, U_2, \dots, U_n\}$ be a finite collection of open sets. Let $x \in \bigcap_{k=1}^{n} U_k$, then $x \in U_k$ for all $1 \leq k \leq n$, each U_k is open so there exists $\delta_k > 0$ such that $V_{\delta_k}(x) \subseteq U_k$, Now, let $\delta = \min\{\delta_1, \delta_2, \dots, \delta_n\}$, this exists because the set is finite. Since the set is finite, $\delta \in \{\delta_1, \dots \delta_n\}$, thus $\delta > 0$. $V_\delta(x) \subseteq V_{\delta_k}(x)$ for each k since $\delta \leq \delta_k$ for each k. Thus, $V_\delta(x) \subseteq U_k$ for each k and $V_\delta(x) \subseteq \bigcap_{k=1}^{n} U_k$ so $\bigcap_{k=1}^{n} U_k$ is open.

$\square$

Definition : Topological Space

A topological space is a pair (X, τ) of a set X and a collection τ of subsets of X (these are the opens sets) satisfying:

1. Arbitrary union of open sets is open
2. Finite intersection of open sets is open
3. X and $\emptyset$ are open

Section 5.1

Definition : (5.1)

A set $U \subseteq \mathbb{R}$ is open if for all $x \in U$, $\exists\, \delta > 0$ such that $(x - \delta, x + \delta) \subseteq U$. That is, if $\forall\, x \in U$, $\exists\, V_\delta(x) \subseteq U$

This is equivalent to saying $U \subseteq \mathbb{R}$ is open if $\forall\, x \in U$, $\exists\, \delta > 0$ such that if $|x - y| < \delta$, then $y \in U$

(i) $\mathbb{R}$ is open, any $\delta > 0$ works

(ii) $\emptyset$ is open, there are no $x \in \emptyset$

(iii) (a, b) is open, let $\delta = \min\{x - a, b - x\}$

(iv) (a, ∞), $(-\infty, b)$ are both open, let $\delta = x - a$ or $\delta = b - x$

(v) $[a, b]$ is not open

Proposition : (5.3)

(i) If $\{U_\alpha\}$ is any collection of open sets, then $\bigcup\limits_{\alpha} U_\alpha$ is also an open set

(ii) If $\{U_1, U_2, \ldots, U_n\}$ is a finite collection of open sets, then $\bigcap\limits_{k=1}^{n} U_k$ is also an open set

Theorem : (5.5)

Every open set is a countable union of disjoint open intervals

Proof. Let A be an open set. Each $x \in A$ is contained in some interval $(x - \delta, x + \delta)$. There may be a larger open interval containing x, let I_x be this interval. Formally. $I_x = (\alpha, \beta)$ where $\alpha = \inf\{a : (a, x) \subseteq A\}$ and $\beta = \sup\{b : (x, b) \subseteq A\}$. Note that given any pair x, y we have either $I_x = I_y$ or $I_x \cap I_y = \emptyset$. This is because if you had two overlapping intervals that were not equal, then you could expand one of them. These I_x intervals are the ones we are looking for. Note that since each $x \in A$ is in $I_x \subseteq A$, the union of all the I_x equals A. These are countably many distinct intervals by the density of $\mathbb{Q}$ in $\mathbb{R}$, and every interval contains a rational number, so there cannot be more intervals than rationals. $\square$

Section 5.2

Definition : (5.6)

A set $A \subseteq \mathbb{R}$ is closed if A^c is open.

This is equivalent to saying A is closed if there exists an open set B such that $B^c = A$

(i) $[a, b]$ is closed, we have shown $(-\infty, a) \cup (b, \infty)$ is open and $[a, b] = [(-\infty, a) \cup (b, \infty)]^c$

(ii) $\mathbb{R}$ is closed, we have shown $\emptyset$ and $\mathbb{R} = \emptyset^c$

(iii) $\emptyset$ is closed, we have shown $\mathbb{R}$ and $\emptyset = \mathbb{R}^c$

Definition : (5.8)

A point x is a limit point of a set A if there is a sequence of points $a_1, a_2, a_3, \dots$ from $A \setminus \{x\}$ such that $a_n \to x$

A point x is a limit of point A iff every ε-neighbourhood of x intersects A at some point other than x

Theorem : (5.10)

A set A is closed iff it contains all of its limit points

Fact : (5.11)

$$\left(\bigcup_\alpha U_\alpha\right)^c = \bigcap_\alpha U_\alpha^c \quad \text{and} \quad \left(\bigcap_\alpha U_\alpha\right)^c = \bigcup_\alpha U_\alpha^c$$

Proposition : (5.12)

(i) if $\{U_1, U_2, U_3, \dots\}$ is a collection of closed sets, then $\bigcup_{k=1}^n U_k$ is also closed

(ii) if $\{U_\alpha\}$ is a collection of closed sets, then $\bigcap_\alpha U_\alpha$ is also closed

Topology

Theorem

Every open set is a countable union of disjoint open intervals

Proof. Let $A \subseteq \mathbb{R}$ be open. Then, each x is contained in some $V_\delta(x) \subseteq A$. Let I_x be the largest open interval that contains x. Let the upper and lower bounds of these intervals be $\alpha = \inf\{a : (a, x) \subseteq A\}$, $\beta = \sup\{b : (x, b) \subseteq A\}$. Our claim is $I_x = (\alpha, \beta)$.

Let $x, y \in A$, what can we say about I_x and I_y? They could be disjoint ($I_x \cap I_y = \emptyset$). What if they are not disjoint? α would have to be the same for I_x and I_y, likewise for β (otherwise you would contradict how α and β are defined.

Consider $\mathbb{Q}$ (which we know is countable). Consider $A \cap \mathbb{Q}$, this is countable. The collection $\{I_x : x \in A \cap \mathbb{Q}\}$ covers all of A ($\bigcup_{x \in A \cap \mathbb{Q}} I_x = A$) since $\mathbb{Q}$ is dense in $\mathbb{R}$. These must be countable because $A \cap \mathbb{Q}$ is countable.

What is this really saying? We can partition A into disjoint open intervals that are indexed by the rational elements of A. There are countably many rationals in A. What about the irrationals in A? Let $y \in A$ and $y \notin \mathbb{Q}$, A is open by assumption, so there is some $\delta > 0$ such that $y \in V_\delta(y) = (y - \delta, y + \delta) \subseteq A$, and this open interval contains rationals. So there is $x \in A \cap \mathbb{Q}$ that is an element of $V_\delta(y)$.

Thus I_x and I_y are not disjoint because we have an open interval that contains both x and y. That is, $V_\delta(y) \subseteq I_y$ and $V_\delta(y) \subseteq I_x$, thus $I_x = I_y$. $\square$

Note: When we say we have a countable collection of sets, we mean the number of sets is countable, not that the total number of elements is countable.

Example

Consider the collection $\{(n, n+1) : n \in \mathbb{N}\} = \{(1, 2), (2, 3)\}$. This is a countable collection. $\bigcup_{n \in \mathbb{N}} (n, n+1)$ is a countable union with uncountably many elements.

Definition

A set $A \subseteq \mathbb{R}$ is closed if the complement A^c is open.

A set A is closed iff there is an open set B such that $B^c = A$

(i) $\mathbb{R}, \emptyset$ are open $\Rightarrow \emptyset, \mathbb{R}$ are closed

(ii) Rationals & Irrationals

- $\mathbb{Q}$ is not open $\Rightarrow \mathbb{R} \setminus \mathbb{Q}$ are not closed
- Could $\mathbb{Q}$ be closed? Equivalently, is $\mathbb{R} \setminus \mathbb{Q}$ open?
- Let $x \in \mathbb{R} \setminus \mathbb{Q}$, any δ-neighbourhood of x is $(x - \delta, x + \delta)$ contains rationals and irrationals, so $(x - \delta, x + \delta)$ is never a subset of $\mathbb{R} \setminus \mathbb{Q}$
- Thus, $\mathbb{R} \setminus \mathbb{Q}$ is not open and $\mathbb{Q}$ is not closed

(iii) Closed Intervals : $A = [a, b]$, $\Rightarrow A^c = (-\infty, a) \cup (b, \infty)$ is open since the union of open intervals is open

How are $(0, 1)$ and $[0, 1]$ different?

- We could have a sequence of elements from $(0, 1)$ that converge to something outside of $(0, 1)$
- Ex. $a_n = \frac{1}{n+1}$, $a_n \to 0 \notin (0, 1)$
- We will see that 0 is a limit point of $(0, 1)$

Topology

Definition

A point x is a limit point of A if there is a sequence of points $a_1, a_2, \dots$ from $A \setminus \{x\}$ such that $a_n \to x$

Example

Let $A = (0, 1)$ and $a_n = \frac{1}{n+1}$. So, $a_1 = \frac{1}{2}, a_2 = \frac{1}{3}, \dots$. Each $a_n \in A$ and $a_n \to 0$. Therefore, 0 is a limit point of A.

1 is also a limit point of A ($a_n = 1 - \frac{1}{n+1}$, $a_n \in A$, $a_n \to 1$)

It turns out that the set of limit points of A is $[0, 1]$

Example

Let $A = \mathbb{N} = \{1, 2, \dots\}$. Is $n = 1$ a limit point? We need to find a sequence (a_n) where each $a_n \in \mathbb{N} \setminus \{1\}$, but $a_n \to 1$. Recall the definition of convergence of a sequence : $\lim_{n\to\infty} a_n = L$ if for all $\varepsilon > 0$, $\exists\ N \in \mathbb{R}$ such that $|a_n - L| < \varepsilon$ when $n > N$. Consider $\varepsilon = \frac{1}{2}, L = 1$, $|a_n - 1| < \frac{1}{2} \Leftrightarrow a_n \left(\frac{1}{2}, \frac{3}{2}\right)$. We also need $a_n \neq 1$ for all n. So this is not possible and thus 1 is not a limit point. None of the elements of $\mathbb{N}$ are limit points of $\mathbb{N}$.

What about other reals, could they be limit points? Can we find a sequence (a_n) where each $a_n \in \mathbb{N}$ and $a_n \to x$? We can find $\varepsilon > 0$ where $(x - \varepsilon, x + \varepsilon)$ doesn't contain any naturals. So no, the naturals have no limit points.

Example : (Exercise 5.7)

x is a limit point of A iff every ε-neighbourhood of x intersects A at some point other than x

Proof. Let x be a limit point of A, let $\varepsilon > 0$. By definition of limit point, there exists a sequence (a_n) in $A \setminus \{x\}$ that converges to x. For $\varepsilon > 0$, $\exists\ N \in \mathbb{R}$ such that $|a_n - x| < \varepsilon$ for $n > N$. Let $m \in \mathbb{N}$ such that $m > N$, thus $a_m \neq x$, $a_m \in A$ and $a_m \in (x - \varepsilon, x + \varepsilon)$. ε was arbitrary, so every ε-neighbourhood of x intersects A at some point other than x.

Suppose every ε-neighbourhood of x intersects A at some point other than x. To prove x is a limit point, we must find a sequence (a_n) in $A \setminus \{x\}$ that converges to x. Consider $\varepsilon = \frac{1}{n} > 0$, $\left(x - \frac{1}{n}, 1 + \frac{1}{n}\right)$ contains at least one element from $A \setminus \{x\}$ (by assumption). For each $n \in \mathbb{N}$, let $a_n \in \left(x - \frac{1}{n}, x + \frac{1}{n}\right)$ where $a_n \in A \setminus \{x\}$. Thus, $x - \frac{1}{n} \leq a_n \leq x + \frac{1}{n}$ for each n, We can apply the squeeze theorem ($x - \frac{1}{n} \to x$, $x + \frac{1}{n} \to x$) thus $a_n \to x$. Therefore x is a limit point of A. □

Theorem

A set A is closed iff it contains it's limit points

Proof. See the proof in the textbook. □

Example

What are the limit points of $\mathbb{Q}$?

---

x is a limit point of $\mathbb{Q}$ iff every ε-neighbourhood of x contains elements from $\mathbb{Q}$ other than x. $(x-\varepsilon, x+\varepsilon)$ contains rationals and irrationals. If $x \in \mathbb{Q}$, can we find $y \in (x-\varepsilon, x+\varepsilon)$ such that $y \neq x$ but $y \in \mathbb{Q}$. The density property of reals says for $a, b \in \mathbb{R}$ with $a < b$, then there exists $x \in \mathbb{Q}$ with $a < c < b$. (We could also prove that between any two reals there is a rational). Therefore, any $x \in \mathbb{R}$ is a limit point of $\mathbb{Q}$.

Topology

DeMorgan's Law for Sets

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

This can be generalized to:

$$\left(\bigcup_{\alpha \in I} U_\alpha\right)^c = \bigcap_{\alpha \in I} U_\alpha^c$$
$$\left(\bigcap_{\alpha \in I} U_\alpha\right)^c = \bigcup_{\alpha \in I} U_\alpha^c$$

Proposition

1. If $\{U_1, U_2, \ldots, U_n\}$ are closed sets, then $\bigcup_{k=1}^{\infty} U_k$ is also closed
2. If $\{U_\alpha\}_{\alpha \in I}$ is a collection of closed sets, then $\bigcap_{\alpha \in I} U_\alpha$ is closed

That is, we can do finite unions and arbitrary intersections of closed sets to get a closed set. This follows from the fact that we can do arbitrary unions and finite intersections of open sets to get an open set.

Definition : Covers

Let A be a set. The set $\{U_\alpha\}_{\alpha \in S}$ is a cover of A if $A \subseteq \bigcup_{\alpha \in S} U_\alpha$

If each U_α is open, we say it is an open cover

If $\{U_\alpha\}_{\alpha \in S}$ has a finite subset $\{U_\alpha\}_{\alpha \in F}$ (assuming $F \subseteq S$) which is still a cover of A, then $\{U_\alpha\}_{\alpha \in F}$ is a finite subcover

Example

Consider the set $A = \{1, 2, 3\}$. A cover (not open however) of this set would be:

$$\{(-3, 0), (-1, 2), \{5\}, [1, 4]\}$$

since

$$A \subseteq (-3, 0) \cup (-1, 2) \cup \{5\} \cup [1, 4] = (-3, 4] \cup \{5\}$$

Example

Let $A = \mathbb{R}$, $U_z = \{z - 1.5, z + 1.5\}$ for $z \in \mathbb{Z}$, then $\{U_z\}_{z \in \mathbb{Z}}$ is an open cover of A. Does it have a finite subcover?

Suppose there is a finite subcover. Since it is finite, let $z_1, z_2, \ldots, z_k$ be the integers that correspond to the finite subcover $\{U_{z_i}\}_{z_i \in \{z_1, z_2, \ldots, z_k\}}$

Any finite set of numbers has a maximal element, let z_M be the max of $\{z_1, z_2, \ldots, z_k\}$. When we do the union of the sets in the subcover, we are bounded above by $z_M + 1.5$ since the sets in the subcover are of the form $(z - 1.5, z + 1.5)$

It is not possible since $\mathbb{R}$ is not bounded above. Therefore, this cover does not have a finite subcover

Example

Consider $A = (0, 4)$, $B = [0, 4]$. $\left\{\left(\frac{1}{k}, 4 - \frac{1}{k}\right)\right\}_{k=1}^{\infty}$ is an open cover for A, and $\left\{\left(\frac{1}{k}, 4 - \frac{1}{k}\right)\right\}_{k=1}^{\infty} \cup (-0.1, 0.1) \cup (3.9, 4.1)$ is an open cover for B. Do either of these open covers have a finite subcover?

For A, any finite subset of the cover corresponds to some max M such that the union is $\left(\frac{1}{M}, 4 - \frac{1}{M}\right)$, but there exists $x \in \mathbb{R}$ such that $0 < x < \frac{1}{M}$, so any finite subset could not cover all of A.

The open cover for B does have an open subcover. We will see this next class.

Open Covers

Example : Continuing from last time

An open cover of $[0,4]$ is:

$$\left\{\frac{1}{k}, 4-\frac{1}{k}\right\}_{k=1}^{\infty} \cup (-0.1, 0.1) \cup (3.9, 4.1)$$

Find a finite subcover.

We know that when $k=1$, we get $(1,3)$, for $k=2$, we get $\left(\frac{1}{2}, 3.5\right)$, for $k=3$, we have $\left(\frac{1}{3}, 3.\overline{66}\right)$, etc. We want to find an interval that intersects with our other two, $(-0.1, 1)$ and $(3.9, 4.1)$.

$$\begin{aligned} \frac{1}{k} &< 0.1 \\ k &> 10 \end{aligned} \qquad \begin{aligned} 4-\frac{1}{k} &> 3.9 \\ -\frac{1}{k} &> -0.1 \\ \frac{1}{k} &< 0.1 \\ k &> 10 \end{aligned}$$

Consider $k=11$, then

$$\left\{\frac{1}{k}, 4-\frac{1}{k}\right\}_{k=1}^{11} \cup (-0.1, 0.1) \cup (3.9, 4.1)$$

is a finite subcover of $[0,4]$.

Note: $\left(\frac{1}{11}, 4-\frac{1}{11}\right) \cup (-0.1, 0.1) \cup (3.9, 4.1)$ is also a finite subcover

We can remove some sets the cover, but we can't change the actual sets. For example, $(-0.1, 4.1)$ covers $[0,4]$ but it is not one of our original sets.

1 Compact Sets

Definition

A set A is compact if every open cover of A contains a finite subcover.

For example, $(0,4)$ is not compact, but $[0,4]$ is.

Theorem : Heine-Borel

A set $S \subseteq \mathbb{R}$ is compact iff it is closed and bounded.

Proof. Assume S is compact

Consider the open cover $\{I_n\}$ with $n \in \mathbb{N}$ and $I_n = (-n, n)$. $S \subseteq \mathbb{R}$ and $\bigcup_{n\in\mathbb{N}} I_n = \mathbb{R}$, thus it is an open cover of S. S is compact, so there is a finite subcover. The finite subcover is $\{I_{n_1}, I_{n_2}, \ldots, I_{n_k}\}$ where $n_1 < n_2 < \cdots < n_k$ and each $n_i \in \mathbb{N}$. So, $I_{n_1} \subseteq I_{n_2} \subseteq \cdots \subseteq I_{n_k}$

since $n_1 < n_2 < \cdots < n_k$ (nested). $\bigcup_{i=1}^{k} I_{n_i} = I_{n_k} = (-n_k, n_k)$ so $S \subseteq (-n_k, n_k)$. That is, for all $x \in S$, $-n_k \leq x \leq n_k$ (in fact, $-n_k < x < n_k$), so S is bounded.

Suppose (toward contradiction) that S is not closed. Thus, S does not contain all of its limit points. There exists $x \notin S$ and a sequence (a_n) with each $a_n \in S \setminus \{x\} = S$ (we know $x \notin S$) and $a_n \to x$. For each $n \in \mathbb{N}$, let $U_n = \left(-\infty, x - \frac{1}{n}\right) \cup \left(x + \frac{1}{n}, \infty\right)$. $\bigcup_{n\in\mathbb{N}} U_n = \mathbb{R} \setminus \{x\}$ and $S \subseteq \mathbb{R} \setminus \{x\}$ so this is an open cover of S. S is compact, so this cover has a finite subcover given by $\{U_{n_1}, U_{n_2}, \ldots, U_{n_k}\}$ where $n_1 < n_2 < \cdots < n_k$. $\bigcup_{i=1}^{k} U_{n_i} = U_{n_k} = \left(-\infty, x - \frac{1}{n_k}\right) \cup \left(x + \frac{1}{n_k}, \infty\right)$. $\left(x - \frac{1}{n_k}, x + \frac{1}{n_k}\right)$ is disjoint from this finite subcover of S, thus it is disjoint from S. Let $\varepsilon = \frac{1}{n_k} > 0$, then for all n, any sequence (a_n) with each $a_n \in S$ is such that $a_n \notin \left(x - \frac{1}{n_k}, x + \frac{1}{n_k}\right)$. $|a_n - x| \geq \varepsilon$, thus (a_n) cannot converge to x, thus x is not a limit point. This is a contradiction which came from assuming S did not contain all of its limit points, hence S does contain all of its limit points, therefore S is closed.

We have proven if S is compact, then it is closed and bounded

We will show the other direction of the proof next time. □

Example

Show that the Cantor Set is closed and bounded.

We know that $C_0 = [0, 1]$, $C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$, etc.

$$C = \bigcap_{n=0}^{\infty} C_n$$

We notice that $C_0 \supseteq C_1 \supseteq \cdots \supseteq C$, and that C_0 is bounded, so C is also bounded.

Also, C is an arbitrary intersection of closed sets, and we know that is closed.

Therefore, C is both closed and bounded.

Continuing our proof

Proof. Last time, we proved that if S is compact, then it is closed and bounded.

Assume S is closed and bounded. Let $\mathcal{F}$ be an open cover of S. We need to show there is a finite subcover. For each $x \in S$, define $S_x = S \cap (\infty, x]$ (elements of S that are $\leq x$). S is bounded by assumption so $\inf(S)$ and $\sup(S)$ exist (by completeness of $\mathbb{R}$). Recall proposition 3.29, which states there are sequences in $\mathbb{R}$ that converge to $\inf(S)$ and to $\sup(S)$ (2 separate sequences since limits are unique). Also, S is closed, so it contains its limit points.

Let $L = \inf(S)$ and $M = \sup(S)$, $L, M \in S$ (since S is closed). $S_L = \{y \in S : y \leq L\} = \{L\}$. $\mathcal{F}$ covers all of S, and $L \in S$, so there is some $U \in \mathcal{F}$ such that $L \in U$. We thus have a finite subcover of $S_L = \{L\}$, namely U. Let $B = \{x : \mathcal{F}$ has a finite subcover of $S_x\}$. We know $L \in B$ since S_L has a finite subcover.

We want to show that we can get all elements from S in B so that F has a finite subcover (and hence S is compact). $M = \sup(S)$, is $M \in B$? If $M \in B$, $S_M = \{y \in S : y \leq M\} = S$, we are done because $S_M = S$ would have a finite subcover. Suppose $M \notin B$, each $S_x \subseteq S$, so $B \subseteq S$. B is bounded since S is bounded. B is a nonempty subset of $\mathbb{R}$ that is bounded above, so $\sup(B)$ exists. Let $T = \sup(B)$. There are two cases: $T \in S$ or $T \notin S$.

If $T \in S$: $\mathcal{F}$ covers S so there is some $U \in \mathcal{F}$ that covers T. U is open so there is $\delta > 0$ such that $V_\delta(T) = (T - \delta, T + \delta) \subseteq U$. Idea: the finite subcover that includes U covers all elements in $(T - \delta, T + \delta)$. In particular, $\left(T - \delta, T + \frac{\delta}{2}\right] \subseteq U$. $T - \delta < T = \sup(B)$ so $T - \delta \in B$. There is a finite subcover $\{F_1, F_2, \ldots, F_k\}$ from $\mathcal{F}$ that covers $S \cap (-\infty, T - \delta\}$. Thus, $\{F_1, F_2, \ldots, F_k\} \cup U$ covers $S \cap \left(-\infty, T + \frac{\delta}{2}\right]$. $T + \frac{\delta}{2} \in B$, this contradicts that $T = \sup(B)$.

If $T \notin S$: Idea - increase T a bit and not change the set, so $S_T = S_{T+\varepsilon}$ for some $\varepsilon > 0$. If $T \notin S$, S is closed so S^c is open. There exists $\delta > 0$ such that $(T - \delta, T + \delta) = V_\delta(T) \subseteq S^c$, so disjoint from S. So $\left[T - \frac{\delta}{2}, T + \frac{\delta}{2}\right] \subseteq S^c \Rightarrow \left[T - \frac{\delta}{2}, T + \frac{\delta}{2}\right] \cap S = \emptyset$. That is, there are no elements of S in $\left[T - \frac{\delta}{2}, T + \frac{\delta}{2}\right]$. $S \cap \left(-\infty, T - \frac{\delta}{2}\right] = S \cap \left(-\infty, T + \frac{\delta}{2}\right]$ because no elements of S are in $\left[T - \frac{\delta}{2}, T + \frac{\delta}{2}\right]$. Since $T - \frac{\delta}{2} < T = \sup(B)$, there is a finite subcover $\{F_1, F_2, \ldots, F_k\}$ that covers $S \cap \left(-\infty, T - \frac{\delta}{2}\right]$ so it also covers $S \cap \left(-\infty, T + \frac{\delta}{2}\right]$, thus $T + \frac{\delta}{2} \in B$ which contradicts that $T = \sup(B)$.

Therefore, we must have $\sup(B) = M$, which means S has a finite subcover, and so S is compact. $\square$

Topology

Example : (Exercise 5.22)

Let A be the set of numbers in $[0,1]$ whose decimal expansions use only the numbers 2, 5, and 8. Prove that A is closed (and hence compact).

The complement of $[0,1]$ is $(-\infty,0)\cup(0,\infty)$. Let this set be B_0.

$$B_1 = (-\infty,0.2)\cup(0.3,0.5)\cup(0.6,0.8)\cup(0.9,\infty)$$

B_1 is the set of numbers that have atleast one digit that is not 2, 5 or 8.

$$\begin{aligned} B_2 = (-\infty,0.2)&\cup(0.2,0.22)\cup(0.23,0.25)\cup(0.26,0.28)\cup(0.29,0.5)\\ &\cup(0.5,0.52)\cup(0.53,0.55)\cup(0.56,0.58)\cup(0.59,0.8)\\ &\cup(0.8,0.82)\cup(0.83,0.85)\cup(0.86,0.88)\cup(0.89,\infty)\end{aligned}$$

Keep going, $\bigcup_{n\geq 0} B_n$ will be the union of open sets whose complement is A, so A is closed.

Continuity

Definition : Continuity

A function $f : A \to \mathbb{R}$ is continuous at a point $c \in A$ if for all $\varepsilon > 0$, there exists some $\delta > 0$ such that for all $x \in A$ with $|x-c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$

$$|x-c| < \delta \Leftrightarrow x \in (c-\delta, c+\delta) = V_\delta(c)$$

$$|f(x) - f(c)| < \varepsilon \Leftrightarrow f(x) \in (f(c)-\varepsilon, f(c)+\varepsilon) = V_\varepsilon(f(c))$$

If f is continuous at every point in the domain, we say f is continuous.

Example

Consider the Dirichlet function:

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Is this function continuous?

Note that any open interval in $\mathbb{R}$ contains both rationals and irrationals. Consider $\varepsilon = \frac{1}{2}$, if $c \in \mathbb{Q}$ we have that $f(c) = 1$. If $|f(x) - f(c)| < \frac{1}{2} \Leftrightarrow f(x) \in \left(\frac{1}{2}, \frac{3}{2}\right)$. For any $\delta > 0$, $(c-\delta, c+\delta)$ contains irrationals, so there is some $x_0 \in (c-\delta, c+\delta)$ with $x_0 \notin \mathbb{Q}$ so $f(x_0) = 0$. f is not continuous at c since $f(x_0) \notin \left(\frac{1}{2}, \frac{3}{2}\right)$. Similarly $f(x)$ is not continuous at any irrational points.

Therefore, the Dirichlet function is continuous nowhere.

Note : Modified Dirichlet function

The modified Dirichlet function, defined by

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

is only continuous at the point $x = 0$.

Theorem

The following are equivalent:

(i) f is continuous at c

(ii) $\forall\, \varepsilon > 0, \exists\, \delta > 0$ such that $\forall\, x \in A$ with $|x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$

(iii) For any ε-neighbourhood of $f(c)$ denoted by $V_\delta(f(c))$, there exists some δ-neighbourhood of c denoted by $V_\delta(c)$ with the property that $\forall\, x \in A$ with $x \in V_\delta(c)$, we have $f(x) \in V_\varepsilon(f(c))$

(iv) For all sequences (a_n) from A which converge to c ($\lim_{n\to\infty} a_n = c$) we have $(f(a_n))$ converging to $f(c)$. That is, $\lim_{n\to\infty} f(a_n) = f(\lim_{n\to\infty} a_n)$

(v) $\lim_{x\to c} f(x) = f(c)$

Topological Continuity

Definition

Let $X, Y \subseteq \mathbb{R}$, $f : X \to Y$, let $B \subseteq Y$. The preimage of B, denoted by $f^{-1}(B)$ is:

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \subseteq X$$

Example

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x + 3$, $B = (0, 4)$. Find the preimage of B.

$$\begin{aligned} f^{-1}(B) &= \{x \in \mathbb{R} : f(x) \in [0, 4]\} \\ &= (-3, 1) \end{aligned}$$

Preimage

Recall that the preimage of B under f is defined as

$$f^{-1}(B) = \{x \in X : f(x) \in B\}$$

Examples:

1. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x + 3$, $B = (0, 4)$
   $f^{-1}(B) = (-3, 1)$
2. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x + 3$, $B = \mathbb{R}$
   $f^{-1}(B) = \mathbb{R}$
3. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, $B = (4, 9)$
   $f^{-1}(B) = (-3, 2) \cup (2, 3)$
4. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, $B = (-3, 1) \cup (4, 9)$
   $f^{-1}(B) = (-3, -2) \cup (-1, 1) \cup (2, 3)$
5. $f : [-1, 5] \to \mathbb{R}$, $f(x) = x + 3$, $B = (0, 4)$
   $f^{-1}(B) = [-1, 1)$
   This is not open in $\mathbb{R}$, but we could express it as $[-1, 5] \cap (-3, 1)$. That is, the intersection of the domain and some open set.

Theorem

Let $f : X \to \mathbb{R}$, then f is continuous if and only if for every open set $B \subseteq \mathbb{R}$, $f^{-1}(B) = A \cap X$ for some open set A.

Proof. The proof will be covered in the next lecture. □

Topological Continuity

Theorem

Let $f : X \to \mathbb{R}$, then f is continuous if and only if for every open set B, $f^{-1}(B) = A \cap X$ for some open set A.

Proof. Assume f is continuous and $B \subseteq \mathbb{R}$ with B open. Let $x_0 \in f^{-1}(B)$, so $f(x_0) \in B$. B is open and $f(x_0) \in B$ so there exists $\varepsilon > 0$ such that $V_\varepsilon(f(x_0)) \subseteq B$. f is continuous, so there exists $\delta > 0$ such that for all $x \in X$ with $x \in V_\delta(x_0)$ we have $f(x) \in V_\varepsilon(f(x_0))$. We have $f(x) \in V_\varepsilon(f(x_0)) \subseteq B$, thus $x \in f^{-1}(B)$. If $X = \mathbb{R}$, then we can express $f^{-1}(B)$ as $\bigcup_{x' \in f^{-1}(B)} V_{\delta'}(x')$ where δ' is such that $V_{\delta'}(x') \subseteq f^{-1}(B)$. The union of open sets is open, so f^{-1} is open. In general, if $f : X \to \mathbb{R}$, $f^{-1}(B) = \bigcup_{x' \in f^{-1}(B)} (V_{\delta'}(x') \cap X) = (V_{\delta'}(x')) \cap X = A \cap X$ where A is open.

Now assume for every open set B, $f^{-1}(B) = A \cap X$ for some open set A. Let $x_0 \in X$ and $\varepsilon > 0$. $V_\varepsilon(f(x_0))$ is an open set. By assumption, $f^{-1}(V_\varepsilon(f(x_0))) = A \cap X$ for some open set A and we have $x_0 \in f^{-1}(V_\varepsilon(f(x_0)))$ since $f(x_0) \in V_\varepsilon(f(x_0))$ thus $x_0 \in A \cap X$, so $x_0 \in A$ and A is open so there exists $\delta > 0$ such that $x_0 \in V_\delta(x_0) \subseteq A$. Thus if $x \in X$ and $x \in V_\delta(x_0)$ we have $f(x) \in V_\varepsilon(f(x_0))$. Hence, f satisfies the topological continuity equivalence. □

Extreme Value Theorem (EVT)

Theorem : Extreme Value Theorem

Suppose $f : X \to \mathbb{R}$ is continuous. If $A \subseteq X$ is compact, then $f(A)$ is compact.

Proof. Assume $f : X \to \mathbb{R}$ is continuous and $A \subseteq X$ is compact. We want to show that $f(A)$ is compact. Let $\{U_\alpha\}$ be an open cover of $f(A)$. Each $f^{-1}(U_\alpha) = V_\alpha \cap X$ for some open set V_α by the previous theorem. We also know $f^{-1} \subseteq V_\alpha$. Consider $\{V_\alpha\}$ is a collection of open sets. Our claim is that this set covers A. First, we will show that $\{f^{-1}(U_\alpha)\}$ covers A (but it is not necessarily open). Let $x_0 \in A$, consider $f(x_0) \in f(A)$. There is some U_{α_0} that contains $f(x_0)$ because we have an open cover, thus $x_0 \in f^{-1}(U_{\alpha_0})$ and $\{f^{-1}(U_\alpha)\}$ covers all of A. Each $f^{-1}(U_\alpha) \subseteq V_\alpha$ so $\{U_\alpha\}$ also covers A and is an open cover. $\{V_\alpha\}$ is an open cover, so it has a finite subcover $\{V_1, V_2, \ldots, V_n\}$ corresponding to $\{f^{-1}(U_1), \ldots, f^{-1}(U_n)\}$ which still covers A. Thus, $\{U_1, \ldots, U_n\}$ covers $f(A)$, this is a finite subcover, so $f(A)$ is therefore compact. □

Example

Consider $f : (0,1) \to \mathbb{R}$, $f(x) = \frac{1}{x}$. Let $A = [0.2, 0.4]$. Show $f(A)$ is compact.

$f(A) = [2.5, 5]$. This is compact, and it has a max and min. If instead we did $f((0,1)) = (1, \infty)$, we would see this set has an inf, but no min and it is not bounded above.

Extreme Value Theorem

Theorem

Suppose $f : X \to \mathbb{R}$ is continuous. If $A \subseteq X$ is compact, then $f(A)$ is compact.

Corollary

The image of a compact set under a continuous function is bounded (and also closed) by Heine Borel

Example

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 5$, $A = [2, 4]$. Show that the image of A is compact, find the sup and inf, and determine if they are limit points.

First, we will show that f is continuous. Let $\varepsilon > 0$, given $c \in \mathbb{R}$, we have $V_\varepsilon(f(c)) = V_\varepsilon(5) = (5 - \varepsilon, 5 + \varepsilon)$. We need $\delta > 0$ such that for $x \in \mathbb{R}$ with $x \in V_\delta(x)$ we have $f(x) \in V_\varepsilon(5)$. $f(x) = 5$ for all x, let $\delta = 1$, then for $x \in V_1(c) = (c - 1, c + 1)$ we have $f(x) = 5 \in V_\varepsilon(5)$. Note that any $\delta > 0$ would work.

$f(A) = \{f(x) : x \in A\} = \{5\}$ is compact. We can see that $\sup(f(A)) = \inf(f(A)) = 5$.

Recall: A point x is a limit point of a set A if there is a sequence of points $a_1, a_2, \ldots$ from $A \setminus \{x\}$ such that $a_n \to x$. Also, a point x is a limit point of a set A iff every ε-neighbourhood of x intersects A at some point other than x.

$V_\varepsilon(5) = (5 - \varepsilon, 5 + \varepsilon)$ does this contain some other point of $f(A)$ other than 5? No since $A \setminus \{x\} = \emptyset$

Proposition : (3.29)

Suppose $S \subseteq \mathbb{R}$ is bounded above. Then, there exists a sequence (a_n) where $a_n \in S$ for all n and $\lim_{n\to\infty} a_n = \sup(S)$

Likewise if S is bounded below, there is a sequence (b_n) where each $b_n \in S$ and $b_n \to \inf(S)$

Example

Give an example of a set with exactly one limit point.

Let $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}$. Each $\frac{1}{n}$ is not a limit point, look at $V_\varepsilon\left(\frac{1}{n}\right) \subseteq \left(\frac{1}{n+1}, \frac{1}{n-1}\right)$ for $n > 1$, and $V_\varepsilon(1) = \left(1 - \frac{1}{4}, 1 + \frac{1}{4}\right)$. 0 is a limit point since $V_\delta(0) = (-\delta, \delta)$ contains some $\frac{1}{n}$.

Theorem : Extreme Value Theorem (EVT)

A continuous function on a compact set attains its sup and inf. That is, there is some a_0, b_0 in the set such that $f(a_0) = \sup$ of the image, $f(b_0) = \inf$ of the image.

Proof. Let A be compact, and f be continuous. $f(A)$ is also compact by the previous theorem. Thus, $f(A)$ is closed and bounded by the Heine Borel theorem. Let $M = \sup(\{f(a) : a \in A\})$, $L = (\{f(a) : a \in A\})$. These exist by completeness of the reals. By proposition 3.29, there is a sequence $f(a_n)$ converging to M and a sequence $f(b_n)$ converging to L. $f(A)$ is closed so they converge to some $f(a_0)$ and $f(b_0)$ inside $f(A)$ where $a_n \to a_0$ and $b_n \to b_0$ (by continuity). We know $f(a_n) \to M$, and $f(A)$ is closed. If $M \notin f(A)$, since $(f(A))^c$ is open, then there is some $\delta > 0$ such that $M \in (M - \delta, M + \delta) \subseteq (f(A))^c$. But then, $M - \frac{\delta}{2} < M$ is also an upper bound for $f(A)$ which contradicts that M is the least upper bound. Thus $M \in f(A)$. Likewise, $L \in f(A)$.

Where did continuity come in? We used the sequential form of continuity from theorem 6.17: f is continuous at $c \in A \Leftrightarrow$ for all sequences (a_n) in A which converge to c, we have $f(a_n) \to f(c)$. (We had $f(a_n) \to M = f(a_0)$, $f(b_n) \to L = f(b_0)$ with $a_n \to a_0$ and $b_n \to b_0$ $\square$

Midterm Practice Problems

Example

Prove $f(x) = x^2$ is continuous at $x = 3$.

$f(x)$ is continuous at c is for every $V_\varepsilon(f(c))$ there is some $V_\delta(c)$ such that for all $x \in X$ with $x \in V_\delta(c)$ we have $f(x) \in V_\varepsilon(f(c))$. In this case, we have $X = \mathbb{R}$, $c = 3$, and $f(c) = 3^2 = 9$. Let $\varepsilon > 0$, $V_\varepsilon(9) = (9-\varepsilon, 9+\varepsilon)$. We want to find $\delta > 0$ such that for $x \in V_\delta(3) = (3-\delta, 3+\delta)$ we have $f(x) \in V_\varepsilon(9)$. $f(x) = x^2 \in (9-\varepsilon, 9+\varepsilon) \Leftrightarrow |x^2 - 9| < \varepsilon$. $|x^2 - 9| = |x+3| \cdot |x-3|$. Assume $\delta \leq 1$ so $|x-3| < \delta = 1 \Leftrightarrow x \in (2,4)$, which gives an upper bound for $|x+3| < 7 = 4+3$. Now, $|x^2 - 9| = |x+3| \cdot |x-3| < 7|x-3|$. Let $\delta = \min\left\{\frac{\varepsilon}{7}, 1\right\}$. Then, $|x^2 - 9| < 7 \cdot \frac{\varepsilon}{7}$ when $|x-3| < \delta$. This is equivalent to saying $f(x) \in V_\varepsilon(9)$ when $x \in V_\delta(3)$.

Example : (Exercise 5.27)

Let $U_1, U_2, \ldots, U_n$ be a finite open cover of B (where B is compact). This implies $B \subseteq U_1 \cup U_2 \cup \cdots \cup U_n$. Is it possible that $B = U_1 \cup U_2 \cup \cdots \cup U_3$

B is compact, so it is closed and bounded by Heine Borel. The union of any collection of open sets is open. If $B = U_1 \cup U_2 \cup \cdots \cup U_n$, then B is also open. Is it possible that B is closed and open? B is also bounded, so $\sup(B)$ exists. B is closed since it is compact, so $\sup(B) \in B$.

If B is open and $\sup(B) \in B$, there exists $\delta > 0$ such that $V_\delta(sup(B)) = (\sup(B) - \delta, \sup(B) + \delta) \subseteq B$. Consider $\sup(B) + \frac{\delta}{2} \in V_\delta(\sup(B))$ but $\sup(B) + \frac{\delta}{2} > \sup(B)$ since $\delta > 0$. This contradicts the definition of the supremum. So, B cannot be open. Thus, this is not possible.

Definition : Interior

The interior of a set A, denoted by $\text{Int}(A)$, is the set of points x such that there is an open neighbourhood of x that is a subset of A.

Example

What is the interior of $A = (0, 4]$?

$4 \notin \text{Int}(A)$ since $V_\delta(4) = (4-\delta, 4+\delta)$ contains elements bigger than 4. What if $x \in (0,4)$? We know $(0,4)$ is open, so by definition each $x \in (0,4)$ is in some $V_\delta(x) \subseteq (0,4)$. Therefore, $\text{Int}(A) = (0,4)$

Definition : Exterior

The exterior of a set A, denoted by $\text{Ext}(A)$, is the set of points x such that there is an open neighbourhood of x that is a subset of A^c. That is, $\text{Ext}(A) = \text{Int}(A^c)$.

Example

What is the exterior of $A = (0, 4]$?

$A^c = (\infty, 0] \cup (4, \infty)$. $\text{Int}\,(A^c) = \text{Ext}\,(A) = (-\infty, 0) \cup (4, \infty)$.

Definition : Boundary

The boundary of a set A, denoted by ∂A, is the set of points x such that every neighbourhood of x contains points from A and A^c.

Example

What are the boundaries of $A = (0, 4]$ and $B = [0, 4]$?

$\partial A = \{0, 4\}$ and $\partial B = \{0, 4\}$.

Note

We could show that for any set A,

$$\mathbb{R} = \text{Int}\,(A) \cup \partial A \cup \text{Ext}\,(A)$$

where each of these sets are disjoint.

Example

Find the interior, exterior and boundary for $A = \mathbb{Z}$.

$\text{Int}\,(A) = \emptyset$, $\text{Ext}\,(A) = \mathbb{R} \setminus \mathbb{Z}$, $\partial A = \mathbb{Z}$.

466 Projects / Presentations

- Portfolio of good examples
  - Weierstrass Function (continuous everywhere, differentiable nowhere)
  - Hilbert Curve (space filling curve)
  - Peano Curve (space filling curve)
  - Topologist's Sine Curve
- Idea
  - Go into detail
  - Give definitions, prove at least one thing
  - Maybe give a bit of history
  - Look at properties: continuity, compactness, max/min, differentiability
- Resources
  - Curious curves
  - Space filling curves
  - Counterexamples in Topology / Analysis
- Last week of February - decide on an example

Intermediate Value Theorem

Example : Hiker Problem

- Start climbing at 7 AM
- Reach summit at 2 PM
- Stay overnight
- Start descending at 7 AM
- Reach base at 2 PM
- There must be a moment in time when you are at the exact same spot on the mountain at the same time

Lemma

If f is continuous, and $f(c) > 0$, then there exists $\delta > 0$ such that $f(x) > 0$ for all $x \in (c - \delta, c + \delta)$.

Likewise, if $f(c) < 0$, we can find $\delta > 0$ such that for all $x \in (c - \delta, c + \delta)$ we have $f(x) < 0$.

Proof. Let $\varepsilon = \frac{f(c)}{2} > 0$. By continuity, there exists $\delta > 0$ such that for $x \in (c-\delta, c+\delta)$, we have $f(x) \in (f(c) - \varepsilon, f(c) + \varepsilon) = \left(f(c) - \frac{f(c)}{2}, f(c) + \frac{f(c)}{2}\right) = \left(\frac{f(c)}{2}, \frac{3f(c)}{2}\right)$. That is, $f(x) > \frac{f(c)}{2} > 0$, so $f(x) > 0$.

Likewise, if $f(c) < 0$, we would let $\varepsilon = \frac{|f(c)|}{2}$. □

Proposition

If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have different signs (one positive, one negative), then there is some $c \in (a, b)$ for which $f(c) = 0$.

Proof. Without loss of generality, suppose $f(a) > 0$, $f(b) < 0$.

Let $A = \{t : f(x) > 0 \; \forall \, x \in [a, t]\}$. We have $a \in A$ since $[a, a] = \{a\}$ and we know $f(a) > 0$. We also know $b \notin A$ since $f(b) < 0$. So, $A \subseteq [a, b]$, A is bounded above by b so by completeness of $\mathbb{R}$, $\sup(A)$ exists. Let $c = \sup(A)$, thus $c \in (a, b)$. If $f(c) = 0$ then we are done. If $f(c) \neq 0$ then $f(c) > 0$ or $f(c) < 0$.

If $f(c) > 0$ then by the previous lemma, there exists $\delta > 0$ such that for all $x \in (c-\delta, c+\delta)$ we have $f(x) > 0$. Is $c - \delta \in A$? For all values of $x \in A$, $f(x) > 0$, so $c - \delta \in A$ because we need $f(x) > 0$ for any $x < c$ otherwise we could have a smaller upper bound for A. Thus, $f(x) > 0$ for all $x \in [a, c+\delta) = [a, c-\delta] \cup (c-\delta, c+\delta)$. We have $f\left(c - \frac{\delta}{2}\right) > 0$ and $c + \frac{\delta}{2} \in A$, which contradicts that $c = \sup(A)$ since $c + \frac{\delta}{2} > c$ and c was an upper bound. The contradiction came from $f(c) > 0$. Similarly, we would get a contradiction if $f(c) < 0$.

Therefore, $f(c) = 0$ □

Intermediate Value Theorem

Theorem : Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and α is any number between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ for which $f(c) = \alpha$

Proof. Either $f(a) < \alpha < f(b)$ or $f(b) < \alpha < f(a)$. Without loss of generality assume $f(a) < \alpha < f(b)$.

Let $g(x) = f(x) - \alpha$, by continuity laws, $g(x)$ is also continuous on $[a, b]$.

$$g(a) = f(a) - \alpha < 0 \qquad \text{since } f(a) < \alpha$$

$$g(b) = f(b) - \alpha > 0 \qquad \text{since } f(b) > \alpha$$

By the previous theorem, there exists $c \in (a, b)$ such that $g(c) = 0$. Therefore $f(c) - \alpha = 0 \Rightarrow f(c) = \alpha$ □

Example : (6.46

et f, g be continuous on $[a, b]$. Suppose $f(a) < g(a)$ and $f(b) > g(b)$, prove there exists $c \in (a, b)$ such that $f(c) = g(c)$.

Let $h(x) = g(x) - f(x)$, h is also continuous by continuity laws

$$h(a) = g(a) - f(a) > 0$$

$$h(b) = g(b) - f(b) < 0$$

By the Intermediate Value Theorem (with $\alpha = 0$) there exists a $c \in (a, b)$ with $h(c) = 0$. Thus $g(c) - f(c) = 0$, so $g(c) = f(c)$.

Uniform Continuity

Definition : Continuity

$f : X \to \mathbb{R}$ is continuous at c if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ when $x \in X$ and $|x - c| < \delta$

Example

Show that $f(x) = x^2$ is continuous on $[0, 4]$

$|f(x) - f(c)| = |x^2 - c^2| = |x + c| \cdot |x - c|$. We can find an upper bound for $|x + c|$ since we are restricted to $[0, 4]$. We know $c \in [0, 4]$ and $x \in [0, 4]$, so $|x + c| \leq 20$ for all $x, c \in [0, 4]$ (we could use 8 instead of 20). So $|x^2 - c^2| \leq 20 \cdot |x - c|$. Let $\varepsilon > 0$, $\delta = \frac{\varepsilon}{20}$. Then,

$$\begin{aligned} |x^2 - c^2| \leq 20 \cdot |x - c| &< 20\delta \\ &= 20 \cdot \frac{\varepsilon}{20} = \varepsilon \end{aligned}$$

Definition : Uniform Continuity

$f : A \to \mathbb{R}$ is uniformly continuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$.

How is this different to normal continuity?

- δ has to work for all $x, y \in A$, it can't depend on c
- For continuity we found an upper bound for $|x + c|$ on $[0, 4]$
- $|f(x) - f(c)|$ is going to grow faster than $|x - c|$ if we are on an unbounded domain like $(0, \infty)$

Recall: For $f : (0, \infty) \to \mathbb{R}$, $f(x) = x^2$, we still need $|x^2 - c^2| = |x + c| \cdot |x - c|$

- For continuity, suppose $\delta \leq 1$, $|x - c| < 1 \Rightarrow |x + c| < 2c + 1$
- $\delta = \min\left\{1, \frac{\varepsilon}{2c+1}\right\}$ (δ depends on c)
- Now if $\varepsilon > 0$ and $x, y \in (0, \infty)$, can we find $\delta > 0$ that always works?
- Idea - Can find x, y with $|x - y| < \delta$ for any δ but $|x^2 - y^2 = |x + y| \cdot |x - y|$, can get x, y big enough so $|x^2 - y^2| \geq \varepsilon$

Uniform Continuity

Definition : Uniform Continuity

$f : A \to \mathbb{R}$ is uniformly continuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in A$, if $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$

Compare with the definition of continuity:

Definition : Continuity

f is continuous at some limit point c if for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - c| < \delta$ then $|f(x) - f(y)| < \varepsilon$

See examples from last class and videos in the notebook (under course resources)

Example : C

nsider $f(x) = x^2$. Is this uniformly continuous?

$f(x)$ is uniformly continuous on $[0, 1]$, but not uniformly continuous on $(0, \infty)$

On $[0, 1]$ there is an upper bound to the steepness of the curve $\frac{|f(x)-f(y)}{|x-y|}$. There is no upper bound to this quantity on $(0, \infty)$, that is the curve gets steeper and steeper.

Proposition

If $f : A \to \mathbb{R}$ is continuous and A is compact, then f is uniformly continuous on A

Proof. Let $\varepsilon > 0$, then $\frac{\varepsilon}{2} > 0$. For each $c \in A$, there exists $\delta(c) > 0$ for which if $x \in A$ and $|x - c| < \delta(c)$ then $|f(x) - f(c)| < \frac{\varepsilon}{2}$ since f is continuous

$\delta(c) > 0$ so $\frac{\delta(c)}{2} > 0$. For each $c \in A$, the open interval $\left(c - \frac{\delta(c)}{2}, c + \frac{\delta(c)}{2}\right)$ contains c

Consider the collection $\left\{\left(c - \frac{\delta(c)}{2}, c + \frac{\delta(c)}{2}\right) : c \in A\right\}$. This is an open cover of A, every $c \in A$ is covered by $\left(c - \frac{\delta(c)}{2}, c + \frac{\delta(c)}{2}\right)$. A is compact, so this has a finite subcover.

We can express the finite subcover as $\left\{\left(c_i - \frac{\delta(c_i)}{2}, c_i + \frac{\delta(c_i)}{2}\right) : 1 \leq i \leq n\right\}$

Let $\delta = \min\left\{\frac{\delta(c_1)}{2}, \frac{\delta(c_n)}{2}, \ldots, \frac{\delta(c_n)}{2}\right\}$, the set is finite so the minimum exists and it is an element of the set (it is some $\frac{\delta(c_k)}{2}$)

Let $x, y \in A$ and $|x - y| < \delta$. $x \in \left(c_j - \frac{\delta(c_j)}{2}, c_j + \frac{\delta(c_j)}{2}\right)$ for some $j \Leftrightarrow |x - c_j| < \frac{\delta(c_j)}{2}$

We also have $|x - y| < \delta \leq \frac{\delta(c_j)}{2}$

$|x - c_j| + |x - y| = |c_j - x| + |x - y| < \frac{\delta(c_j)}{2} + \frac{\delta(c_j)}{2} = \delta(c_j)$

$|c_j - x + x - y| \leq |c_j - x| + |x - y|$ by triangle inequality

so $|c_j - y| < \delta(c_j)$

$$\begin{aligned}
|f(x) - f(y)| &= |f(x) - f(c_j) + f(c_j) - f(y)| \\
&\leq |f(x) - f(c_j)| + |f(y) - f(c_j)| \qquad \text{by triangle inequality} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \qquad \text{because of how we defined } \delta(c_j) \\
&= \varepsilon
\end{aligned}$$

□

Graphical Interpretations of Slope and Tangent

We are skipping this section since it was all covered in first year calculus.

The Derivative

Definition : Differentiablility

Let I be an interval, $f : I \to \mathbb{R}$, and $c \in I$. We say f is differentiable at $c \in I$ if the following limit exists

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

If C is a collection of points where f is differentiable, then the derivative of f is a function $f' : C \to \mathbb{R}$, where

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

Continuity and Differentiability

Theorem : S

ppose I is an interval and $f : I \to \mathbb{R}$ is differentiable at $c \in I$. Then, f is continuous at c.

Proof. $f'(c)$ exists, so let $L = \lim_{x\to c} \frac{f(x)-f(c)}{x-c}$. It suffices to show $\lim_{x\to c} f(x) = f(c) \Leftrightarrow \lim_{x\to c} f(x) - f(c) = 0$. In the limit $x \to c$, we have $x \neq c$ so $x - c \neq 0$.

$$\begin{aligned}
\lim_{x\to c} f(x) - f(c) &= \lim_{x\to c} (f(x) - f(c)) \left(\frac{x-c}{x-c}\right) && \text{since } x - c \neq 0 \\
&= \left[\lim_{x\to c} \frac{f(x)-f(c)}{x-c}\right] \cdot \left[\lim_{x\to c} x - c\right] && \text{by limit laws} \\
&= L \cdot 0 = 0
\end{aligned}$$

Thus f is continuous at c □

The converse is not true however, we can have a function that is continuous at c, but not differentiable

Example

Show that $f(x) = |x|$ is not differentiable at $c = 0$ (we know it is continuous at $c = 0$)

Let's try to find $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$ for $c = 0$

$$\begin{aligned}
\lim_{x\to 0} \frac{|x| - 0}{x - 0} &= \lim_{x\to 0} \frac{|x|}{x} \\
&= \begin{cases} \lim_{x\to 0} \frac{x}{x} & \text{if } x \geq 0 \\ \lim_{x\to 0} \frac{-x}{x} & \text{if } x < 0 \end{cases} \\
&= \begin{cases} \lim_{x\to 0} 1 & \text{if } x \geq 0 \\ \lim_{x\to 0} -1 & \text{if } x < 0 \end{cases}
\end{aligned}$$

The limits don't agree as $x \to 0$, so our function is not differentiable at $c = 0$

Note: $|x|$ is continuous on $\mathbb{R}$ and differentiable for all reals except 0

Differentiation Rules

Let I be an interval, $k \in \mathbb{R}$, and $f, g : I \to \mathbb{R}$ where both f and g are differentiable at $c \in \mathbb{R}$

Proposition

1. $(f+g)'(c) = f'(c) + g'(c)$
2. $(f-g)'(c) = f'(c) - g'(c)$
3. $(k \cdot f)'(c) = k \cdot f'(c)$

Theorem : Product Rule

$(fg)'(c) = f'(c) \cdot g(c) + f(c) \cdot g'(c)$

Proof. We need to use the definition of the derivative

$$\begin{aligned}
\lim_{x \to c} \frac{(fg)(x) - (fg)(c)}{x - c} &= \lim_{x \to c} \frac{f(x)\,g(x) - f(c)\,g(c)}{x - c} \\
&= \lim_{x \to c} \frac{f(x)\,g(x) - f(c)\,g(x) + f(c)\,g(x) - f(c)\,g(c)}{x - c} \\
&= \lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) g(x) + \lim_{x \to c} f(c) \left(\frac{g(x) - g(c)}{x - c} \right) \\
&= f'(c) \cdot g(c) + f(c) \cdot g'(c)
\end{aligned}$$

□

Theorem : Chain Rule

Let I_1, I_2 be intervals, $g : I_1 \to I_2$, $f : I_2 \to \mathbb{R}$, $(f \circ g) : I_1 \to \mathbb{R}$, g is differentiable at $c \in I_1$, f is differentiable at $g(c) \in I_2$, then $f \circ g$ is differentiable at c and $(f \circ g)' = f'(g(c)) \cdot g'(c)$

Proof. Careful:

$$\lim_{x \to c} \frac{f(g(x)) - f(g(c))}{x - c} = \lim_{x \to c} \frac{f(g(x)) - f(g(c))}{g(x) - g(c)} \cdot \frac{g(x) - g(c)}{x - c}$$

In the limit, we know $x \to c$ so $x \neq c$. However, we cannot assume $g(x) \neq g(c)$

Define a new function

$$d(y) = \begin{cases} f(y) - f(g(c)) & y \neq g(c) \\ f'(g(c)) & y = g(c) \end{cases}$$

This is well defined because we are given that f is differentiable at $x = g(c)$, so $f'(g(c))$ exists

If $g(x) \neq g(c)$ then $\frac{f(g(x)) - f(g(c))}{x - c}$ can be multiplied by $\frac{g(x) - g(c)}{g(x) - g(c)}$ so that

$$\frac{f(g(x)) - f(g(c))}{x - c} = \frac{f(g(x)) - f(g(c))}{g(x) - g(c)} \cdot \frac{g(x) - g(c)}{x - c}$$

If $g(x) = g(c)$, then $g(x) - g(c) = 0$

$$\frac{f(g(x)) - f(g(c))}{x - c} = \frac{f(g(c)) - f(g(c))}{x - c} = \frac{0}{x - c} = 0 \qquad \text{in lim since } x - c \neq 0$$

In either case we have

$$\begin{aligned}\lim_{x\to c}\frac{f\left(g\left(x\right)\right)-f\left(g\left(c\right)\right)}{x-c} &= \lim_{x\to c} d\left(g\left(x\right)\right)\left[\frac{g\left(x\right)-g\left(c\right)}{x-c}\right]\\ &= \lim_{x\to c} d\left(g\left(x\right)\right)\cdot\lim_{x\to c}\frac{g\left(x\right)-g\left(c\right)}{x-c}\\ &= f'\left(g\left(c\right)\right)\cdot g'\left(c\right)\end{aligned}$$

$\square$

Topoligcal Sine Curve

$$g_n(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$$g_0(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$$g_1(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

See the book for more information

Local Maximum and Minimum

Definition

$f : A \to \mathbb{R}$ has a local max at $c \in A$i f there exists $\delta > 0$ such that for all $x \in A$ with $|x - c| < \delta$, we have $f(x) \leq f(c)$

Likewise, f has a local min at c if there exists $\delta > 0$ such that for all $x \in A$ with $|x - c| < \delta$ we have $f(x) \geq f(c)$

Note

Recall: A sequence (S_n) where each term $S_n \geq 0$ and $S_n \to L$, then $L \geq 0$

Proposition

Let I be an open interval and f is differentiable at $c \in I$. If f has a local max or min at c, then $f'(c) = 0$

Proof. f is differentiable at c, so $f'(c) = \frac{f(x)-f(c)}{x-c}$ exists. Suppose there is a local max at c. Let $\delta > 0$ be such that for all $x \in I$ with $|x - c| < \delta$ we have $f(x) \leq f(c)$

$$\frac{f(x) - f(c)}{x - c} \quad \text{is} \quad \begin{cases} \geq 0 & \text{if } x > c \\ \leq 0 & \text{if } x < c \end{cases}$$

We can find a sequence (a_n) from I with $a_n \to c$ and $\frac{f(a_n)-f(c)}{a_n-c} \to f'(c)$. Idea - Find two different sequences, one is from the left and the other is from the right

(a_n) is a sequence in I with $c - \delta < a_n < c$ for all n and $a_n \to c$, so $\frac{f(a_n)-f(c)}{a_n-c} \geq 0$ for all n. Each $S_n \geq 0$ and $S_n \to f'(c)$, thus $f'(c) \geq 0$

(b_n) is a sequence in I with $c < b_n < c + \delta$ for all n and $b_n \to c$, so $\frac{f(b_n)-f(c)}{b_n-c} \leq 0$ for all n. Each $t_n \leq 0$ and $t_n \to f'(c)$, thus $f'(c) \leq 0$

We have $f'(c) \geq 0$ and $f'(c) \leq 0$ so $f'(c) = 0$.

Likewise if f has a local min at c $\square$

Note : Intermediate Value Property (IVP)

A function has the intermediate value property on $[a, b]$ if when α is any value between $f(a)$ and $f(b)$ there is some $c \in (a, b)$ with $f(c) = \alpha$

We have proven that if f is continuous on $[a, b]$, then it has the IVP (This was the IVT)

Example

Does f has to be continuous? Find a function f on $[0, 4]$, $f(0) = 0$, $f(4) = 0$, which has the IVP, but is not continuous.

There are many functions that would work here

$$f(x) = \begin{cases} 4x & x \leq 1 \\ 0 & -1 < x < 4 \\ 4 & x = 4 \end{cases}$$

Given a function f, if f is differentiable on some set A, then f' is defined on A. There is no guarantee that f' is a continuous function, however we can prove f' has the IVP.

Theorem : Darboux's Theorem

Let f be differentiable on $[a, b]$. If α is between $f'(a)$ and $f'(b)$ then there exists $c \in (a, b)$ where $f'(c) = \alpha$

Proof. Without loss of generality assume $f'(b) < \alpha < f'(a)$. Define $g(x) = f(x) - \alpha x$. Then g is differentiable and $g'(x) = f'(x) - \alpha$. We want $f'(c) = \alpha$

$$g'(a) = f'(a) - \alpha > 0$$

$$g'(b) = f'(b) - \alpha < 0$$

$[a, b]$ is compact and g is continuous (since g is differentiable) so by the extreme value theorem (EVT) g attains a max. Could it be at an endpoint?

Consider $x = a$. If $g(x) \leq g(a)$ for all $x \in [a, b]$ then $\frac{g(x)-g(a)}{x-a} \leq 0$ for $x \in (a, b]$ but $g'(a) > 0$ so can't have a max at a. Likewise can't be at b.

There is a max at some $c \in (a, b)$. Let $\delta = \min\{c - a, b - c\}$ we could have $g(x) \leq g(c)$ for $x \in [a, b]$ with $|x - c| < \delta$. Thus it is also a local max. Therefore $g'(c) = 0$ and $f'(c) = \alpha$ $\square$

Mean Value Theorem

Theorem : Rolle's Theorem

$f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$ then there exists $c \in (a, b)$ with $f'(c) = 0$

Proof. By the Extreme Value Theorem, f has a max at some $c_1 \in [a, b]$ and a min at some $c_2 \in [a, b]$. If $c_1 \in (a, b)$ or $c_2 \in (a, b)$ then it is a local max (or min) and $f'(c_1) = 0$ (or $f'(c_2) = 0$)

Now assume they are both endpoints. Without loss of generality assume $c_1 = a$ and $c_2 = b$. The max is $f(a)$, min is $f(b)$ and $f(a) = f(b)$ by assumption. So for any $x \in [a, b]$ we have $f(b) \leq f(x) \leq f(a)$. This forces $f(x) = f(a)$ for all $x \in [a, b]$ (A constant function). It can be shown that the derivative of a constant function is 0 and so for any $c \in (a, b)$ we have $f'(c) = 0$ □

Theorem : Mean Value Theorem (MVT)

$f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). There exists $c \in (a, b)$ with $f'(c) = \frac{f(b)-f(a)}{b-a}$. I.e., $f'(c)(b - a) = f(b) - f(a)$

Proof. Consider the line through $(a, f(a))$ and $(b, f(b))$. This gives us the equation

$$y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a)$$

$$y = \left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a) = S(x)$$

Let $g(x) = f(x) - S(x)$ (The difference between our curve and the secant line). Both f and S are continuous on $[a, b]$ and differentiable on (a, b) so g is also continuous on $[a, b]$ and differentiable on (a, b).

$$g(a) = f(a) - \left[\left(\frac{f(b) - f(a)}{b - a}\right)(a - a) + f(a)\right] = f(a) - f(a) = 0$$

$$g(b) = f(b) - \left[\left(\frac{f(b) - f(a)}{b - a}\right)(b - a) + f(a)\right] = f(b) - f(b) = 0$$

So $g(a) = g(b)$. We can apply Rolle's theorem. There exists a $c \in (a, b)$ with $g'(c) = 0$.

$$g(x) = f(x) - S(x) \Rightarrow g'(x) = f'(x) - S'(x) = f'(x) - \left(\frac{f(b) - f(a)}{b - a}\right)$$

So $g'(c) = 0 \Rightarrow f'(c) = \frac{f(b)-f(a)}{b-a}$ as desired. □

Corollary

Let I be an interval and $f : I \to \mathbb{R}$ is differentiable. If $f'(x) = 0$ for all $x \in I$ then f is constant.

Proof. Let $x_1, x_2 \in I$ with $x_1 < x_2$. f is differentiable on $[x_1, x_2]$ (thus continuous) so there exists $c \in (x_1, x_2)$ with $f'(c) = \frac{f(x_2)-f(x_1)}{x_2-x_1} = 0$ since $c \in I$. $f(x_2) - f(x_1) = 0 \Rightarrow f(x_1) = f(x_2)$. x_1, x_2 were arbitrary so f must be constant on I. □

Corollary

Let I be an interval, $f, g : I \to \mathbb{R}$, both differentiable on I. If $f'(x) = g'(x)$ for all $x \in I$ then $f(x) = g(x) + C$ for some $c \in \mathbb{R}$.

Proof. Let $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x)$. Then apply the previous corollary. □

Corollary

Let I be an interval, $f : I \to \mathbb{R}$, f is differentiable on I

(i) f is monotonic increasing iff $f'(x) \geq 0$ for all $x \in I$

(ii) f is monotonic decreasing iff $f'(x) \leq 0$ for all $x \in I$

Proof. Assume f is increasing. So for all $x_1, x_2 \in I$ with $x_1 < x_2$ we have $f(x_1) \leq f(x_2)$. $\frac{f(x)-f(c)}{x-c}$ is either $\frac{(-)}{(-)}$ or $\frac{(+)}{(+)}$ or 0. $f'(c) = \lim_{x \to c} \frac{f(x)-f(c)}{x-c} \geq 0$

Now assume $f'(x) \geq 0$ on I. Let $x_1, x_2 \in I$ with $x_1 < x_2$. $\frac{f(x_2)-f(x_1)}{x_2-x_1} = f'(c) \geq 0$ for some $c \in (x_1, x_2)$. Thus $f(x_2) \geq f(x_1)$. These were arbitrary so f is increasing.

Similar for decreasing. □